

Question number	Answer	Mark
6(a)(i)	C glucose and urea	(1)

Question number	Answer	Mark
6(a)(ii)	A basement membrane	(1)

Question number	Answer	Mark
6(a)(iii)	C sodium co-transport	(1)

Question number	Answer	Mark
6(a)(iv)	An explanation that includes any two of the following points: • because the concentration of urea is greater in the filtrate than in the blood (1) • so urea will diffuse down its concentration gradient (1) • because the walls of the proximal tubule and capillaries are permeable to urea (1)	(2)

Question number	Answer	Mark
6(b)(i)	• correct figures selected from the table and subtracted (1) • percentage decrease calculated (1) Example of calculation: $7.0 - 0.6 = 6.4$ $(6.4 \div 7) \times 100 = 91.43$	(2)

Question number	Answer	Mark
6(b)(ii)	B at 1-minute intervals between 5 minutes and 15 minutes	(1)

Question number	Answer	Mark
6(b)(iii)	An explanation that includes any four of the following points: • because the carotid artery will carry the blood straight up to the hypothalamus (1) • which will detect the lower solute potential of the blood (1) • as a result, the (posterior) pituitary gland will release ADH (1) • ADH will result in more reabsorption of water from the {distal tubules / collecting ducts} so less urine produced (1) • because when solute potential returns to normal, ADH will no longer be	(4)